

1. Connect your laptop and a friend's phone to the same Wi-Fi network, then use the ping command to check if both devices can communicate with each other. Hint: On Windows, use 'ping [IP address]' in Command Prompt; on Android, try a ping app or use Termux.

```
Media State . . . . . : Media disconnected
Connection-specific DNS Suffix  . :

C:\Users\maitri>
C:\Users\maitri>ping 192.168.1.88

Pinging 192.168.1.88 with 32 bytes of data:
Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 192.168.1.88:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\Users\maitri>ping google.com

Pinging google.com [2404:6800:4007:805::200e] with 32 bytes of data:
Reply from 2404:6800:4007:805::200e: time=68ms
Reply from 2404:6800:4007:805::200e: time=71ms
Reply from 2404:6800:4007:805::200e: time=70ms
Reply from 2404:6800:4007:805::200e: time=71ms

Ping statistics for 2404:6800:4007:805::200e:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 68ms, Maximum = 71ms, Average = 70ms

C:\Users\maitri>
```

2. Run the traceroute (tracert on Windows or traceroute on Mac/Linux) command to www.youtube.com from your device and note down the number of hops it takes to reach the destination.

```
Microsoft Windows [Version 10.0.19044.7417]
(c) Microsoft Corporation. All rights reserved.

C:\Users\maitri>tracert www.youtube.com

Tracing route to www.youtube.com [2404:6800:4009:801::200e]
over a maximum of 30 hops:

  1  6 ms   7 ms   2 ms   jiofiber.local.html [2409:40c1:5030:26c6:feb0:deff:fe67:72ec]
  2  49 ms  26 ms  29 ms  2405:200:5210:7:3924:af:3:411
  3  40 ms  45 ms 122 ms  2405:200:5210:7:3925::1
  4  *      *      *      Request timed out.
  5  *      *      *      Request timed out.
  6  53 ms  78 ms  29 ms  2405:200:801:b00::fd6
  7  *      *      *      Request timed out.
  8 139 ms  47 ms  49 ms  2404:6800:8281:c0::1
  9  61 ms  51 ms  43 ms  2404:6800:8281:c0::1
 10  42 ms  79 ms  81 ms  2404:6800:8281:c0::1
 11  49 ms  60 ms  43 ms  2001:4860:0:1::7ba8
 12  56 ms  47 ms  35 ms  2001:4860:0:1::15c0
 13  95 ms  70 ms  45 ms  2001:4860:0:1::8769
 14 159 ms 101 ms  45 ms  2001:4860:0:1::11c1
 15  42 ms  33 ms  67 ms  bom07s10-in-x0e.1e100.net [2404:6800:4009:801::200e]

Trace complete.

C:\Users\maitri>
```

3. Disconnect one device from the Wi-Fi, then try pinging it again from your laptop and describe the error message you receive.

```
Select Command Prompt

Media State . . . . . : Media disconnected
Connection-specific DNS Suffix  . :

C:\Users\maitri>
C:\Users\maitri>ping 192.168.1.88

Pinging 192.168.1.88 with 32 bytes of data:
Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 192.168.1.88:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\Users\maitri>ping google.com

Pinging google.com [2404:6800:4007:805::200e] with 32 bytes of data:
Reply from 2404:6800:4007:805::200e: time=68ms
Reply from 2404:6800:4007:805::200e: time=71ms
Reply from 2404:6800:4007:805::200e: time=70ms
Reply from 2404:6800:4007:805::200e: time=71ms

Ping statistics for 2404:6800:4007:805::200e:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 68ms, Maximum = 71ms, Average = 70ms

C:\Users\maitri>
```

4. Create a simple table listing the IP addresses of all devices connected to your home Wi-Fi (use 'arp -a' on Windows), and identify which device belongs to which person or gadget.

```
Command Prompt

Microsoft Windows [Version 10.0.19044.7417]
(c) Microsoft Corporation. All rights reserved.

C:\Users\maitri>arp -a

Interface: 192.168.58.1 --- 0x5
Internet Address      Physical Address      Type
192.168.58.254        00-50-56-e9-1c-bc    dynamic
192.168.58.255        ff-ff-ff-ff-ff-ff    static
224.0.0.22            01-00-5e-00-00-16    static
224.0.0.251           01-00-5e-00-00-fb    static
224.0.0.252           01-00-5e-00-00-fc    static
239.255.255.250       01-00-5e-7f-ff-fa    static
255.255.255.255       ff-ff-ff-ff-ff-ff    static

Interface: 192.168.31.79 --- 0x6
Internet Address      Physical Address      Type
192.168.31.1          fc-b0-de-67-72-ec    dynamic
192.168.31.103        a8-71-16-ab-34-54    dynamic
192.168.31.255        ff-ff-ff-ff-ff-ff    static
224.0.0.22            01-00-5e-00-00-16    static
224.0.0.251           01-00-5e-00-00-fb    static
224.0.0.252           01-00-5e-00-00-fc    static
239.255.255.250       01-00-5e-7f-ff-fa    static
255.255.255.255       ff-ff-ff-ff-ff-ff    static

Interface: 192.168.125.1 --- 0xd
Internet Address      Physical Address      Type
192.168.125.254       00-50-56-e3-18-3b    dynamic
192.168.125.255       ff-ff-ff-ff-ff-ff    static
224.0.0.22            01-00-5e-00-00-16    static
224.0.0.251           01-00-5e-00-00-fb    static
224.0.0.252           01-00-5e-00-00-fc    static
239.255.255.250       01-00-5e-7f-ff-fa    static
255.255.255.255       ff-ff-ff-ff-ff-ff    static

C:\Users\maitri>
```

5. Write a one-page report explaining step-by-step how you set up the network, tested connectivity using ping and traceroute, and what results or issues you observed. Hint: Include screenshots or command outputs if possible.

The purpose of this activity was to set up a home Wi-Fi network and test the network connection using ipconfig, ping, tracert, and arp -a commands.

First, I connected my laptop to my home Wi-Fi network. I opened Command Prompt and ran the ipconfig command. I noted my network details:

- **IPv4 Address:** 192.168.1.18
- **Subnet Mask:** 255.255.255.0
- **Default Gateway:** 192.168.1.1

I used the ping command to test the connection with another device.

Command:

```
ping 192.168.31.131
```

The command showed "Destination host unreachable." This means the target device could not be reached because it was disconnected or not available on the network.

Next, I used the tracert command to check the path to YouTube.

```
tracert www.youtube.com
```

The command completed successfully and reached YouTube in 15 hops. Some hops showed "Request timed out", which is normal because some routers do not reply to traceroute requests.

I ran the following command:

```
arp -a
```

The command displayed the IP addresses and MAC addresses of the devices connected to my network.

Observations

- My laptop connected successfully to the Wi-Fi network.
- The ipconfig command showed my IP address and default gateway.

- The ping command returned "Destination host unreachable" because the other device was not reachable.
- The tracert command reached YouTube in 15 hops.
- The arp -a command displayed the devices available on the network.